2.40 Differential Equations

Worked Solutions 1–46

1 a)

$$\int \frac{1}{y}\,dy = \int 1\,dx \Rightarrow \ln|y| = x + c \Rightarrow y = \pm e^{x+c}$$
$$\Rightarrow y = \pm e^x e^c \Rightarrow y = Ae^x.$$

1 b)

$$\int \frac{1}{y}\,dy = \int \frac{1}{x}\,dx \Rightarrow \ln|y| = \ln|x| + c \Rightarrow y = \pm e^{\ln|x|+c}$$
$$\Rightarrow y = \pm e^{\ln|x|}e^c \Rightarrow y = Ax.$$

1 c)

$$\int \frac{1}{y}\,dy = \int x^{-2}\,dx \Rightarrow \ln|y| = -x^{-1} + c \Rightarrow y = \pm e^{-x^{-1}+c}$$
$$\Rightarrow y = Ae^{-1/x}.$$

1 d)

$$\int \frac{1}{y}\,dy = \int \frac{1}{6}x^{-4}\,dx \Rightarrow \ln|y| = -\frac{1}{18}x^{-3} + c$$
$$\Rightarrow y = \pm e^{-x^{-3}/18+c} \Rightarrow y = Ae^{-1/(18x^3)}.$$

2 a)

$$\int y\,dy = \int x\,dx \Rightarrow \frac{1}{2}y^2 = \frac{1}{2}x^2 + c \Rightarrow y^2 = x^2 + A$$
$$\Rightarrow y = \pm\sqrt{x^2 + A}.$$

2 b)

$$\int y^2\,dy = \int x^2\,dx \Rightarrow \frac{1}{3}y^3 = \frac{1}{3}x^3 + c \Rightarrow y^3 = x^3 + A$$
$$\Rightarrow y = (x^3 + A)^{1/3}.$$

2 c)

$$\int 2y\,dy = \int x^3\,dx \Rightarrow y^2 = \frac{1}{4}x^4 + c$$
$$\Rightarrow y = \pm(\frac{1}{4}x^4 + c)^{1/2}.$$

2 d)

$$\int 3y^3\,dy = \int x^5\,dx \Rightarrow \frac{3}{4}y^4 = \frac{1}{6}x^6 + c$$
$$\Rightarrow y^4 = \frac{2}{9}x^6 + A$$
$$\Rightarrow y = \pm(\frac{2}{9}x^6 + A)^{1/4}.$$

3)

$$\int x^{-4}\,dx = \frac{1}{10}\int \frac{1}{t}\,dt \Rightarrow -\frac{1}{3}x^{-3} = \frac{1}{10}\ln|t| + c$$
$$x = 1,\ t = 1: \quad -\frac{1}{3} = 0 + c \Rightarrow c = -\frac{1}{3}$$
$$\therefore -\frac{1}{3}x^{-3} = \frac{1}{10}\ln|t| - \frac{1}{3} \Rightarrow x^{-3} = 1 - \frac{3}{10}\ln|t|$$
$$\Rightarrow x^3 = \frac{1}{1 - \frac{3}{10}\ln|t|} \Rightarrow x^3 = \frac{10}{10 - 3\ln|t|}$$
$$\Rightarrow x = (\frac{10}{10 - 3\ln|t|})^{1/3}.$$

4)

$$\int x^{1/2}\,dx = \int t\,dt \Rightarrow \frac{2}{3}x^{3/2} = \frac{1}{2}t^2 + c \Rightarrow x^{3/2} = \frac{3}{4}t^2 + A$$
$$\Rightarrow x = (\frac{3}{4}t^2 + A)^{2/3}$$
$$\text{When } x = 4,\ t = 0: \quad 4 = A^{2/3} \Rightarrow A = 8$$
$$\therefore x = (\frac{3}{4}t^2 + 8)^{2/3}$$
$$\text{When } t = 2\sqrt{39}: \quad x = (\frac{3}{4} \times (2\sqrt{39})^2 + 8)^{2/3} = 25.$$

5 a)

$$\int y^{-2}\,dy = \frac{1}{2}\int \frac{1}{x}\,dx \Rightarrow -y^{-1} = \frac{1}{2}\ln|x| + c$$
$$\Rightarrow y^{-1} = A - \frac{1}{2}\ln|x| \Rightarrow y = \frac{1}{A - \frac{1}{2}\ln|x|} \Rightarrow y = \frac{2}{B - \ln|x|}.$$

5 b)

$$\int y^{-2}\,dy = \int x^{-2}\,dx \Rightarrow -y^{-1} = -x^{-1} + c$$
$$\Rightarrow y^{-1} = x^{-1} + A \Rightarrow y = \frac{1}{x^{-1} + A} \Rightarrow y = \frac{x}{1 + Ax}.$$

5 c)

$$\int y^{-3}\,dy = \int \frac{1}{3}x^{-3}\,dx \Rightarrow -\frac{1}{2}y^{-2} = -\frac{1}{6}x^{-2} + c$$
$$\Rightarrow y^{-2} = \frac{1}{3}x^{-2} + A$$
$$\Rightarrow y = \pm(\frac{1}{\frac{1}{3x^2} + A})^{1/2} \Rightarrow y = \pm(\frac{3x^2}{1 + Bx^2})^{1/2}.$$

6 a)

$$\int \frac{1}{y}\,dy = \int x\,dx \Rightarrow \ln|y| = \frac{1}{2}x^2 + c$$
$$\Rightarrow y = \pm e^{\frac{1}{2}x^2 + c} \Rightarrow y = Ae^{\frac{1}{2}x^2}.$$

6 b)

$$\int \frac{1}{y}\,dy = \int x^2\,dx \Rightarrow \ln|y| = \frac{1}{3}x^3 + c$$
$$\Rightarrow y = \pm e^{\frac{1}{3}x^3 + c} \Rightarrow y = Ae^{\frac{1}{3}x^3}.$$

6 c)

$$\int \frac{1}{y}\,dy = \int 4x^3\,dx \Rightarrow \ln|y| = x^4 + c$$
$$\Rightarrow y = \pm e^{x^4 + c} \Rightarrow y = Ae^{x^4}.$$

6 d)

$$\int \frac{1}{y}\,dy = \int x^{1/2}\,dx \Rightarrow \ln|y| = \frac{2}{3}x^{3/2} + c$$
$$\Rightarrow y = \pm e^{\frac{2}{3}x^{3/2} + c} \Rightarrow y = Ae^{\frac{2}{3}x^{3/2}}.$$

7)

$$\int \frac{1}{P}\,dP = \int -100t^3\,dt \Rightarrow \ln|P| = -25t^4 + c$$
$$\Rightarrow P = \pm e^{-25t^4 + c} \Rightarrow P = Ae^{-25t^4}$$

When $t = 0,\ P = 500:\quad 500 = Ae^0 \Rightarrow A = 500$

$$\therefore P = 500e^{-25t^4}$$

When $t = 0.7,\quad P = 500e^{-25\times 0.7^4} = 1.24$ to 3 s.f.

8 a)

$$\int y^{-2}\,dy = \int x\,dx \Rightarrow -y^{-1} = \frac{1}{2}x^2 + c \Rightarrow y^{-1} = A - \frac{1}{2}x^2$$
$$\Rightarrow y = \frac{1}{A - \frac{1}{2}x^2} \Rightarrow y = \frac{2}{B - x^2}.$$

8 b)

$$\int y^{-2}\,dy = \int x^2\,dx \Rightarrow -y^{-1} = \frac{1}{3}x^3 + c$$
$$\Rightarrow y^{-1} = A - \frac{1}{3}x^3 \Rightarrow y = \frac{1}{A - \frac{1}{3}x^3} \Rightarrow y = \frac{3}{B - x^3}.$$

8 c)

$$\int y^{-3}\,dy = \int \frac{2}{3}x^3\,dx \Rightarrow -\frac{1}{2}y^{-2} = \frac{1}{6}x^4 + c$$
$$\Rightarrow y^{-2} = A - \frac{1}{3}x^4$$
$$\Rightarrow y = \pm(\frac{1}{A - \frac{1}{3}x^4})^{1/2} \Rightarrow y = \pm(\frac{3}{B - x^4})^{1/2}.$$

9)

$$\int y^{-3}\,dy = \int \frac{1}{3}x^3\,dx \Rightarrow -\frac{1}{2}y^{-2} = \frac{1}{12}x^4 + c$$
$$x = 1,\ y = 6: \quad -\frac{1}{2} \times 6^{-2} = \frac{1}{12} \times 1^4 + c \Rightarrow c = -\frac{7}{72}$$
$$\therefore -\frac{1}{2}y^{-2} = \frac{1}{12}x^4 - \frac{7}{72} \Rightarrow y^{-2} = \frac{7}{36} - \frac{1}{6}x^4$$
$$\Rightarrow y = \pm(\frac{1}{\frac{7}{36} - \frac{1}{6}x^4})^{1/2}$$
$$\Rightarrow y = \pm(\frac{36}{7 - 6x^4})^{1/2}$$
$$\text{When } x = 0: \quad y = \pm(\frac{36}{7})^{1/2} = \pm\frac{6}{\sqrt{7}} = \pm\frac{6\sqrt{7}}{7}.$$

10 a)

$$\int y\,dy = \int \sin x\,dx \Rightarrow \frac{1}{2}y^2 = -\cos x + c$$
$$\Rightarrow y^2 = A - 2\cos x \Rightarrow y = \pm(A - 2\cos x)^{1/2}.$$

10 b)

$$\int 4y\,dy = \int \cos x\,dx \Rightarrow 2y^2 = \sin x + c$$
$$\Rightarrow y^2 = \frac{1}{2}\sin x + A \Rightarrow y = \pm(\frac{1}{2}\sin x + A)^{1/2}.$$

10 c)

$$\int 3y^{1/2}\,dy = \int \sin 2x\,dx \Rightarrow 2y^{3/2} = -\frac{1}{2}\cos 2x + c$$
$$\Rightarrow y^{3/2} = A - \frac{1}{4}\cos 2x \Rightarrow y = (A - \frac{1}{4}\cos 2x)^{2/3}.$$

10 d)

$$\int 4y^{1/2}\,dy = \int 5\cos 4x\,dx \Rightarrow \frac{8}{3}y^{3/2} = \frac{5}{4}\sin 4x + c$$
$$\Rightarrow y^{3/2} = \frac{15}{32}\sin 4x + A \Rightarrow y = (\frac{15}{32}\sin 4x + A)^{2/3}.$$

11 a)

$$\int y^{1/2}\,dy = \int 8\sqrt{2}\cos 3x\,dx \Rightarrow \frac{2}{3}y^{3/2} = \frac{8\sqrt{2}}{3}\sin 3x + c$$
$$x = 0,\ y = 0: \quad 0 = 0 + c \Rightarrow c = 0$$
$$\therefore \frac{2}{3}y^{3/2} = \frac{8\sqrt{2}}{3}\sin 3x \Rightarrow y^{3/2} = 4\sqrt{2}\sin 3x$$
$$\Rightarrow y = (4\sqrt{2}\sin 3x)^{2/3}.$$

11 b)

A maximum occurs when $\sin 3x = 1$, so $y = (4\sqrt{2})^{2/3} \Rightarrow y = 32^{1/3}$.

12 a)

$$\int e^y\,dy = \int \cos 2x\,dx \Rightarrow e^y = \frac{1}{2}\sin 2x + c$$
$$\Rightarrow y = \ln(\frac{1}{2}\sin 2x + c).$$

12 b)

$$\int \frac{1}{y+1}\,dy = \int \frac{1}{x+1}\,dx \Rightarrow \ln|y+1| = \ln|x+1| + c$$
$$\Rightarrow y + 1 = \pm e^{\ln|x+1|+c} \Rightarrow y = A(x+1) - 1$$
$$\Rightarrow y = Ax + B.$$

12 c)

$$\int e^{0.2y}\,dy = \int 12x^2\,dx \Rightarrow 5e^{0.2y} = 4x^3 + c$$
$$\Rightarrow e^{0.2y} = \frac{4}{5}x^3 + A \Rightarrow 0.2y = \ln(\frac{4}{5}x^3 + A)$$
$$\Rightarrow y = 5\ln(\frac{4}{5}x^3 + A).$$

12 d)

$$\int \frac{1}{4y-3}\,dy = \int e^{-0.2x}\,dx \Rightarrow \frac{1}{4}\ln|4y-3| = -5e^{-0.2x} + c$$
$$\Rightarrow \ln|4y-3| = A - 20e^{-0.2x} \Rightarrow 4y - 3 = e^{A-20e^{0.2x}}$$
$$\Rightarrow 4y = Be^{-20e^{0.2x}} + 3 \Rightarrow y = Me^{-20e^{0.2x}} + \frac{3}{4}.$$

13 a)

$$\frac{dy}{dx} = \frac{e^x}{e^y} \Rightarrow \int e^y\,dy = \int e^x\,dx \Rightarrow e^y = e^x + c$$
$$\Rightarrow y = \ln(e^x + c).$$

13 b)

$$\frac{dy}{dx} = \frac{e^y}{e^x} \Rightarrow \int e^{-y}\,dy = \int e^{-x}\,dx \Rightarrow -e^{-y} = -e^{-x} + c$$
$$\Rightarrow e^{-y} = e^{-x} + A \Rightarrow e^y = \frac{1}{e^{-x}+A}$$
$$\Rightarrow y = \ln(\frac{1}{e^{-x}+A}) \Rightarrow y = \ln(\frac{e^x}{1+Ae^x}).$$

13 c)

$$\frac{dy}{dx} = \frac{2e^{4x}}{e^{3y}} \Rightarrow \int e^{3y}\,dy = \int 2e^{4x}\,dx$$
$$\Rightarrow \frac{1}{3}e^{3y} = \frac{1}{2}e^{4x} + c \Rightarrow e^{3y} = \frac{3}{2}e^{4x} + A$$
$$\Rightarrow 3y = \ln(\frac{3}{2}e^{4x} + A) \Rightarrow y = \frac{1}{3}\ln(\frac{3}{2}e^{4x} + A).$$

13 d)

$$\frac{dy}{dx} = \frac{8e^{4y}}{e^{x/2}} \Rightarrow \int e^{-4y}\,dy = \int 8e^{-x/2}\,dx$$
$$\Rightarrow -\frac{1}{4}e^{-4y} = -16e^{-x/2} + c \Rightarrow e^{-4y} = 64e^{-x/2} + A$$
$$\Rightarrow e^{4y} = \frac{1}{64e^{-x/2}+A} \Rightarrow 4y = \ln(\frac{1}{64e^{-x/2}+A})$$
$$\Rightarrow y = \frac{1}{4}\ln(\frac{1}{64e^{-x/2}+A}) \Rightarrow y = \frac{1}{4}\ln(\frac{e^{x/2}}{64+Ae^{x/2}}).$$

14)

$$\frac{dy}{dx} = \frac{e^y}{e^{6x}} \Rightarrow \int e^{-y}\,dy = \int e^{-6x}\,dx$$

$$\Rightarrow -e^{-y} = -\frac{1}{6}e^{-6x} + c$$

$$x = 0,\ y = 0: \quad -e^0 = -\frac{1}{6}e^0 + c \Rightarrow -1 = -\frac{1}{6} + c$$

$$\Rightarrow c = -\frac{5}{6}$$

$$\Rightarrow -e^{-y} = -\frac{1}{6}e^{-6x} - \frac{5}{6}$$

$$\Rightarrow e^{-y} = \frac{1}{6}e^{-6x} + \frac{5}{6} \Rightarrow e^y = \frac{1}{\frac{1}{6}e^{-6x} + \frac{5}{6}} = \frac{6}{e^{-6x} + 5}$$

$$\Rightarrow y = \ln(\frac{6}{e^{-6x} + 5})$$

$$\text{When } x = 0.5: \quad y = \ln(\frac{6}{e^{-3} + 5}).$$

15)

$$\int 24y^{-1/3}\,dy = \int \frac{4 - x}{x}\,dx \Rightarrow \int 24y^{-1/3}\,dy = \int (\frac{4}{x} - 1)\,dx$$

$$\Rightarrow 36y^{2/3} = 4\ln|x| - x + c \Rightarrow y^{2/3} = \frac{1}{9}\ln|x| - \frac{1}{36}x + A$$

$$\Rightarrow y = \pm(\frac{1}{9}\ln|x| - \frac{1}{36}x + A)^{3/2}$$

$$\Rightarrow y = \pm(\frac{1}{36}(4\ln|x| - x + k))^{3/2}$$

$$\Rightarrow y = \pm(\frac{1}{36})^{3/2}(k - x + 4\ln|x|)^{3/2}$$

$$\Rightarrow y = \pm\frac{(k - x + 4\ln x)^{3/2}}{216} \text{ as } x > 0 \text{ so } |x| = x.$$

16 a)

$$\int \frac{1}{H}\,dH = \int \frac{2}{15}\cos 0.2t\,dt \Rightarrow \ln|H| = \frac{2}{3}\sin 0.2t + c$$

$$t = 0,\ H = 2: \quad \ln 2 = 0 + c \Rightarrow c = \ln 2$$

$$\therefore \ln|H| = \frac{2}{3}\sin 0.2t + \ln 2 \Rightarrow H = \pm e^{\frac{2}{3}\sin 0.2t + \ln 2}$$

$$\Rightarrow H = \pm 2e^{\frac{2}{3}\sin 0.2t}.$$

As H represents the height above the ground, it cannot be negative, so $H = 2e^{\frac{2}{3}\sin 0.2t}$.

16 b)

Maximum height occurs when $\sin 0.2t = 1 \Rightarrow 0.2t = \frac{\pi}{2} \Rightarrow t = \frac{5\pi}{2} \approx 7.85$ seconds.

17)

$$\frac{dy}{dx} = \frac{k(10-x)}{y}$$

$$x=5,\ y=20,\ \frac{dy}{dx}=10: \quad 10 = k \times \frac{10-5}{20} \Rightarrow k = 40$$

$$\Rightarrow \int y\,dy = 40\int (10-x)\,dx$$

$$\Rightarrow \frac{1}{2}y^2 = 40(10x - \frac{1}{2}x^2) + c \Rightarrow y^2 = 800x - 40x^2 + A$$

$$x=5,\ y=20: \quad 20^2 = 800 \times 5 - 40 \times 5^2 + A$$

$$\Rightarrow A = -2600$$

$$\therefore y^2 = 800x - 40x^2 - 2600.$$

18 a)

$$\frac{dr}{dt} = \frac{k}{\sqrt{r}}.$$

where r is the radius in mm, t is the time in seconds, and k is a positive constant.

$$\Rightarrow \int r^{1/2}\,dr = \int k\,dt \Rightarrow \frac{2}{3}r^{3/2} = kt + c$$

$$t=0,\ r=4: \quad \frac{2}{3} \times 4^{3/2} = 0 + c \Rightarrow c = \frac{16}{3}$$

$$\therefore \frac{2}{3}r^{3/2} = kt + \frac{16}{3}$$

$$t=2,\ r=16: \quad \frac{2}{3} \times 16^{3/2} = 2k + \frac{16}{3} \Rightarrow k = \frac{56}{3}$$

$$\therefore \frac{2}{3}r^{3/2} = \frac{56}{3}t + \frac{16}{3} \Rightarrow r^{3/2} = 28t + 8 \Rightarrow r = (28t+8)^{2/3}.$$

18 b)

$r = (28t+8)^{2/3}$ may not be a valid equation to model the ink in the long term as in reality the ink will eventually stop spreading. Whereas in our model the radius grows indefinitely and this is not realistic.

19)

Acceleration is the rate of change of velocity, i.e. $a = \frac{dv}{dt}$.

$$\begin{aligned}
\therefore \frac{dv}{dt} &= 1.02v^3 \Rightarrow \int v^{-3}\,dv = \int 1.02\,dt \\
&\Rightarrow -\frac{1}{2}v^{-2} = 1.02t + c \\
t = 0,\ v = 1: \quad -\frac{1}{2} &= 0 + c \Rightarrow c = -\frac{1}{2} \\
\therefore -\frac{1}{2}v^{-2} &= 1.02t - \frac{1}{2} \Rightarrow v^{-2} = 1 - \frac{51}{25}t \\
&\Rightarrow v^2 = \frac{1}{1 - \frac{51}{25}t} \\
&\Rightarrow v = \pm\sqrt{\frac{1}{1 - \frac{51}{25}t}} \Rightarrow v = \pm\sqrt{\frac{25}{25 - 51t}} \\
&\Rightarrow v = \pm\frac{5}{\sqrt{25 - 51t}}.
\end{aligned}$$

As $v > 0$, $v = \frac{5}{\sqrt{25-51t}}$, as required.

20)

Acceleration is the rate of change of velocity, i.e. $a = \frac{dv}{dt}$.

$$\begin{aligned}
\therefore \frac{dv}{dt} &= -0.02v^2 \Rightarrow \int v^{-2}\,dv = \int -0.02\,dt \\
&\Rightarrow -v^{-1} = -0.02t + c \\
t = 0,\ v = 2: \quad -2^{-1} &= 0 + c \Rightarrow c = -\frac{1}{2} \\
\therefore -v^{-1} &= -0.02t - \frac{1}{2} \Rightarrow v^{-1} = \frac{1}{50}t + \frac{1}{2} \\
&\Rightarrow v = \frac{1}{\frac{1}{50}t + \frac{1}{2}} \Rightarrow v = \frac{50}{t + 25} \text{ as required.}
\end{aligned}$$

21)

Acceleration is the rate of change of velocity, i.e. $a = \frac{dv}{dt}$.

$$\begin{aligned}
\therefore \frac{dv}{dt} &= -1.05v \Rightarrow \int \frac{1}{v}\,dv = \int -1.05\,dt \\
&\Rightarrow \ln|v| = -1.05t + c \Rightarrow v = Ae^{-1.05t} \\
t = 0,\ v = 6: \quad 6 &= Ae^0 \Rightarrow A = 6 \\
\therefore v &= 6e^{-1.05t} \\
\text{When } t = 2, \quad v &= 6e^{-1.05\times 2} = 0.735 \text{ m s}^{-1}.
\end{aligned}$$

22 a)

$$\frac{dT}{dt} = -\frac{k}{\sqrt{T}}.$$

22 b)

The coffee cannot decrease in temperature indefinitely as it will eventually reach room temperature.

23)

Rate **out**: 0.02.

Rate **in**: AV, where A is a positive constant.

$$\frac{dV}{dt} = AV - 0.02 \Rightarrow 50\frac{dV}{dt} = kV - 1.$$

where $k = 50A$, as required.

24 a)

$$\frac{dP}{dt} = kP.$$

24 b)

$$\int \frac{1}{P}\,dP = \int k\,dt \Rightarrow \ln|P| = kt + c \Rightarrow P = Ae^{kt}$$

$$t = 0,\ P = 500: \quad 500 = Ae^0 \Rightarrow A = 500$$

$$\therefore P = 500e^{kt}$$

$$\frac{dP}{dt} = 500ke^{kt}$$

$$t = 0,\ \frac{dP}{dt} = 0.02: \quad 0.02 = 500ke^0 \Rightarrow k = 0.00004$$

$$\therefore P = 500e^{0.00004t}$$

$$\text{When } t = 365, \quad P = 500e^{0.00004\times 365} \approx £507.35.$$

25)

$$\frac{dN}{dt} = kN$$

$$5000 = k \times 60000 \Rightarrow k = \frac{1}{12}$$

$$\therefore \frac{dN}{dt} = \frac{1}{12}N \Rightarrow \frac{dN}{dt} = \frac{N}{12}.$$

26)

$$\frac{dh}{dt} = k\ln h$$

$$0.2 = k\ln 2 \Rightarrow k = \frac{0.2}{\ln 2} = \frac{1}{5\ln 2} = \frac{1}{\ln 32}$$

$$\therefore \frac{dh}{dt} = \frac{1}{\ln 32}\ln h \Rightarrow \frac{dh}{dt} = \frac{\ln h}{\ln 32}.$$

27)

$$V = 6 \times 10 \times d = 60d \Rightarrow d = \frac{1}{60}V$$
$$\frac{dV}{dt} = 0.1 - 0.05d \Rightarrow \frac{dV}{dt} = 0.1 - 0.05 \times (\frac{1}{60}V)$$
$$\Rightarrow \frac{dV}{dt} = \frac{1}{10} - \frac{1}{1200}V$$
$$\Rightarrow \frac{dV}{dt} = \frac{120}{1200} - \frac{V}{1200} \Rightarrow \frac{dV}{dt} = \frac{120 - V}{1200}.$$

28)

$$V = \pi \times 4^2 \times d = 16\pi d \Rightarrow d = \frac{1}{16\pi}V$$
$$\frac{dV}{dt} = 0.8 - 0.2d^2 \Rightarrow \frac{dV}{dt} = 0.8 - 0.2 \times (\frac{1}{16\pi}V)^2$$
$$\Rightarrow \frac{dV}{dt} = \frac{4}{5} - \frac{1}{1280\pi^2}V^2$$
$$\Rightarrow \frac{dV}{dt} = \frac{1024\pi^2}{1280\pi^2} - \frac{V^2}{1280\pi^2}$$
$$\Rightarrow \frac{dV}{dt} = \frac{1024\pi^2 - V^2}{1280\pi^2}$$
$$\Rightarrow \frac{dV}{dt} = \frac{(32\pi - V)(32\pi + V)}{1280\pi^2}.$$

29 a)

$$\int y\,dy = \int (x+2)^3\,dx \Rightarrow \frac{1}{2}y^2 = \frac{1}{4}(x+2)^4 + c$$
$$\Rightarrow y^2 = \frac{1}{2}(x+2)^4 + A \Rightarrow y = \pm(\frac{1}{2}(x+2)^4 + A)^{1/2}.$$

29 b)

$$\int \frac{1}{y}\,dy = \int (2x-5)^4\,dx \Rightarrow \ln|y| = \frac{1}{10}(2x-5)^5 + c$$
$$\Rightarrow y = \pm e^{\frac{1}{10}(2x-5)^5 + c} \Rightarrow y = Ae^{\frac{1}{10}(2x-5)^5}.$$

29 c)

$$\int y^{1/2}\,dy = \int (4x+7)^{1/2}\,dx \Rightarrow \frac{2}{3}y^{3/2} = \frac{1}{6}(4x+7)^{3/2} + c$$
$$\Rightarrow y^{3/2} = \frac{1}{4}(4x+7)^{3/2} + A \Rightarrow y = (\frac{1}{4}(4x+7)^{3/2} + A)^{2/3}.$$

29 d)

$$\int e^y\,dy = \int (3-x)^{2/3}\,dx \Rightarrow e^y = -\frac{3}{5}(3-x)^{5/3} + c$$
$$\Rightarrow y = \ln(c - \frac{3}{5}(3-x)^{5/3}).$$

30 a)

$$\int y\,dy = \int xe^x\,dx$$
$$u = x, \quad \frac{dv}{dx} = e^x$$
$$\frac{du}{dx} = 1, \quad v = e^x$$
$$\therefore \int y\,dy = xe^x - \int e^x\,dx \Rightarrow \frac{1}{2}y^2 = xe^x - e^x + c$$
$$\Rightarrow y^2 = 2xe^x - 2e^x + A \Rightarrow y = \pm(2xe^x - 2e^x + A)^{1/2}.$$

30 b)

$$\int \frac{1}{y}\,dy = \int x\sin 10x\,dx$$
$$u = x, \quad \frac{dv}{dx} = \sin 10x$$
$$\frac{du}{dx} = 1, \quad v = -\frac{1}{10}\cos 10x$$
$$\therefore \int \frac{1}{y}\,dy = -\frac{1}{10}x\cos 10x + \int \frac{1}{10}\cos 10x\,dx$$
$$\Rightarrow \ln|y| = -\frac{1}{10}x\cos 10x + \frac{1}{100}\sin 10x + c$$
$$\Rightarrow y = \pm e^{-\frac{1}{10}x\cos 10x + \frac{1}{100}\sin 10x + c}$$
$$\Rightarrow y = Ae^{-\frac{1}{10}x\cos 10x + \frac{1}{100}\sin 10x}.$$

30 c)

$$\int 20y^2\,dy = \int xe^{2x}\,dx$$
$$u = x, \quad \frac{dv}{dx} = e^{2x}$$
$$\frac{du}{dx} = 1, \quad v = \frac{1}{2}e^{2x}$$
$$\therefore \int 20y^2\,dy = \frac{1}{2}xe^{2x} - \int \frac{1}{2}e^{2x}\,dx$$
$$\Rightarrow \frac{20}{3}y^3 = \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + c$$
$$\Rightarrow y^3 = \frac{3}{40}xe^{2x} - \frac{3}{80}e^{2x} + A$$
$$\Rightarrow y = (\frac{3}{40}xe^{2x} - \frac{3}{80}e^{2x} + A)^{1/3}.$$

30 d)

$$\int 3\ln y\,dy = \int 5x\ln x\,dx.$$

LHS:

$$u = \ln y, \quad \frac{dv}{dy} = 3, \qquad \frac{du}{dy} = \frac{1}{y}, \quad v = 3y.$$

RHS:

$$u = \ln x, \quad \frac{dv}{dx} = 5x, \qquad \frac{du}{dx} = \frac{1}{x}, \quad v = \frac{5}{2}x^2.$$

$$\therefore 3y\ln y - \int 3\,dy = \frac{5}{2}x^2\ln x - \int \frac{5}{2}x\,dx$$

$$\Rightarrow 3y\ln y - 3y = \frac{5}{2}x^2\ln x - \frac{5}{4}x^2 + c.$$

31 a)

$$\frac{8x-10}{(x-3)(x+4)} \equiv \frac{A}{x-3} + \frac{B}{x+4}$$

$$\Rightarrow 8x - 10 \equiv A(x+4) + B(x-3)$$

$$\text{Let } x = 3: \quad 14 = 7A \Rightarrow A = 2$$

$$\text{Let } x = -4: \quad -42 = -7B \Rightarrow B = 6$$

$$\therefore \frac{8x-10}{(x-3)(x+4)} \equiv \frac{2}{x-3} + \frac{6}{x+4}.$$

31 b)

$$\int \frac{1}{y}\,dy = \int \frac{8x-10}{(x-3)(x+4)}\,dx$$

$$\Rightarrow \int \frac{1}{y}\,dy = \int \left(\frac{2}{x-3} + \frac{6}{x+4}\right)dx$$

$$\Rightarrow \ln|y| = 2\ln|x-3| + 6\ln|x+4| + c$$

$$y = 20,\ x = 4: \quad \ln 20 = 0 + 6\ln 8 + c$$

$$\Rightarrow c = \ln 20 - 6\ln 8 \Rightarrow c = \ln 20 - \ln 262144$$

$$\Rightarrow c = \ln\frac{5}{65536}$$

$$\therefore \ln|y| = 2\ln|x-3| + 6\ln|x+4| + \ln\frac{5}{65536}$$

$$\Rightarrow \ln|y| = \ln(x-3)^2 + \ln(x+4)^6 + \ln\frac{5}{65536}$$

$$\Rightarrow \ln|y| = \ln\left(\frac{5(x-3)^2(x+4)^6}{65536}\right)$$

$$\Rightarrow y = \frac{5(x-3)^2(x+4)^6}{65536}.$$

32)

$$\frac{1}{8}\int \frac{1}{y}\,dy = \int \cos x \sin^3 x\,dx \Rightarrow \frac{1}{8}\ln|y| = \frac{1}{4}\sin^4 x + c$$

$$y = 2,\ x = 0: \quad \frac{1}{8}\ln 2 = 0 + c \Rightarrow c = \frac{1}{8}\ln 2$$

$$\therefore \frac{1}{8}\ln|y| = \frac{1}{4}\sin^4 x + \frac{1}{8}\ln 2$$

$$\Rightarrow \ln|y| = 2\sin^4 x + \ln 2 \Rightarrow y = \pm 2e^{2\sin^4 x}.$$

33 a)

$$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dr} = 4\pi r^2$$

$$\frac{dV}{dt} = -k \text{ where } k \text{ is a positive constant}$$

$$\therefore \frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{4\pi r^2} \times -k$$

$$\Rightarrow \frac{dr}{dt} = -(\frac{k}{4\pi})\frac{1}{r^2}$$

$$\Rightarrow \frac{dr}{dt} = -\frac{m}{r^2} \text{ where } m = \frac{k}{4\pi} \text{ as required.}$$

33 b)

$$\int r^2\,dr = \int -m\,dt \Rightarrow \frac{1}{3}r^3 = -mt + c$$

$$t = 0,\ r = 30: \quad \frac{1}{3} \times 30^3 = 0 + c \Rightarrow c = 9000$$

$$\therefore \frac{1}{3}r^3 = -mt + 9000$$

$$t = 2,\ r = 29: \quad \frac{1}{3} \times 29^3 = -2m + 9000 \Rightarrow m = \frac{2611}{6}$$

$$\therefore \frac{1}{3}r^3 = -\frac{2611}{6}t + 9000 \Rightarrow r^3 = -\frac{2611}{2}t + 27000$$

$$\Rightarrow r = (27000 - \frac{2611}{2}t)^{1/3}.$$

34)

Rate **out**: mh^2 where m is a positive constant.

Rate **in**: 2400.

$$\frac{dV}{dt} = 2400 - mh^2$$

$$V = 3600h \Rightarrow \frac{dV}{dh} = 3600$$

$$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt} = 3600 \times \frac{dh}{dt}$$

$$\therefore 3600\frac{dh}{dt} = 2400 - mh^2 \Rightarrow \frac{dh}{dt} = \frac{2}{3} - kh^2.$$

where $k = \frac{m}{3600}$, as required.

35)

$$\int 10P\,dP = \int \sin^2 4t\,dt$$
$$\cos 2A = 1 - 2\sin^2 A$$
$$\text{Let } A = 4t \Rightarrow \cos 8t = 1 - 2\sin^2 4t$$
$$\Rightarrow \sin^2 4t = \frac{1}{2} - \frac{1}{2}\cos 8t$$
$$\therefore \int 10P\,dP = \int (\frac{1}{2} - \frac{1}{2}\cos 8t)\,dt$$
$$\Rightarrow 5P^2 = \frac{1}{2}t - \frac{1}{16}\sin 8t + c$$
$$P = 20,\ t = 0: \quad 5 \times 20^2 = 0 - 0 + c \Rightarrow c = 2000$$
$$\therefore 5P^2 = \frac{1}{2}t - \frac{1}{16}\sin 8t + 2000$$
$$\Rightarrow P^2 = \frac{1}{10}t - \frac{1}{80}\sin 8t + 400$$
$$\Rightarrow P = \pm(\frac{1}{10}t - \frac{1}{80}\sin 8t + 400)^{1/2}$$
$$\Rightarrow P = \pm(\frac{1}{400}(40t - 5\sin 8t + 160000))^{1/2}$$
$$\Rightarrow P = \frac{1}{20}(40t - 5\sin 8t + 160000)^{1/2} \text{ as } P > 0.$$

36 a)

$$\frac{dy}{dx} = \frac{y}{10}(1 - \frac{y}{100}) \Rightarrow \frac{dy}{dx} = \frac{1}{10000} \times y(100 - y)$$
$$\Rightarrow \int \frac{1}{y(100-y)}\,dy = \int \frac{1}{1000}\,dx.$$

LHS:

$$\frac{1}{y(100-y)} \equiv \frac{A}{y} + \frac{B}{100-y} \Rightarrow 1 \equiv A(100-y) + By$$
$$\text{Let } y = 0: \quad 1 = 100A \Rightarrow A = \frac{1}{100}$$
$$\text{Let } y = 100: \quad 1 = 100B \Rightarrow B = \frac{1}{100}$$
$$\therefore \int (\frac{1}{100y} + \frac{1}{100(100-y)})\,dy = \int \frac{1}{1000}\,dx$$
$$\Rightarrow \frac{1}{100}\ln|y| - \frac{1}{100}\ln|100-y| = \frac{1}{1000}x + c$$
$$y = 10,\ x = 0: \quad \frac{1}{100}\ln 10 - \frac{1}{100}\ln 90 = 0 + c \Rightarrow c = \frac{1}{100}\ln\frac{1}{9}$$
$$\therefore \frac{1}{100}\ln y - \frac{1}{100}\ln(100-y) = \frac{1}{1000}x + \frac{1}{100}\ln\frac{1}{9}.$$

as $0 < y < 100$.

$$\Rightarrow \ln(\frac{y}{100-y}) = \frac{1}{10}x + \ln\frac{1}{9} \Rightarrow \frac{y}{100-y} = \frac{1}{9}e^{x/10}$$
$$\Rightarrow y = \frac{100}{9}e^{x/10} - \frac{1}{9}ye^{x/10}$$
$$\Rightarrow y + \frac{1}{9}ye^{x/10} = \frac{100}{9}e^{x/10}$$
$$\Rightarrow y(1 + \frac{1}{9}e^{x/10}) = \frac{100}{9}e^{x/10}$$
$$\Rightarrow y = \frac{(100/9)e^{x/10}}{1 + (1/9)e^{x/10}} \Rightarrow y = \frac{100e^{x/10}}{9 + e^{x/10}}.$$

36 b)

$$y = \frac{100e^{x/10}}{9 + e^{x/10}} \Rightarrow y = \frac{100}{9e^{-x/10} + 1}.$$

As $x \to \infty$, $e^{-x/10} \to 0$, so $y \to \frac{100}{9\times 0+1} = 100$.

37 a)

$$\frac{1}{x(200-x)} \equiv \frac{A}{x} + \frac{B}{200-x} \Rightarrow 1 \equiv A(200-x) + Bx$$
$$\text{Let } x = 0: \quad 1 = 200A \Rightarrow A = \frac{1}{200}$$
$$\text{Let } x = 200: \quad 1 = 200B \Rightarrow B = \frac{1}{200}$$
$$\therefore \frac{1}{x(200-x)} \equiv \frac{1}{200x} + \frac{1}{200(200-x)}.$$

37 b)

$$\int \frac{1}{x(200-x)}\,dx = \int \frac{1}{1000}\,dt$$
$$\Rightarrow \frac{1}{200}\int (\frac{1}{x} + \frac{1}{200-x})\,dx = \int \frac{1}{1000}\,dt$$
$$\Rightarrow \frac{1}{200}\ln|x| - \frac{1}{200}\ln|200-x| = \frac{1}{1000}t + c$$
$$x = 100,\ t = 0: \quad \frac{1}{200}\ln 100 - \frac{1}{200}\ln 100 = 0 + c \Rightarrow c = 0$$
$$\therefore \frac{1}{200}\ln x - \frac{1}{200}\ln(200-x) = \frac{1}{1000}t \text{ as } 0 < x < 200$$
$$\Rightarrow \ln x - \ln(200-x) = \frac{1}{5}t \Rightarrow \ln(\frac{x}{200-x}) = \frac{1}{5}t$$
$$\Rightarrow \frac{x}{200-x} = e^{t/5} \Rightarrow x = 200e^{t/5} - xe^{t/5}$$
$$\Rightarrow x + xe^{t/5} = 200e^{t/5} \Rightarrow x(1 + e^{t/5}) = 200e^{t/5}$$
$$\Rightarrow x = \frac{200e^{t/5}}{1 + e^{t/5}}.$$

37 c)

$$x = \frac{200e^{t/5}}{1+e^{t/5}} \Rightarrow x = \frac{200}{e^{-t/5}+1}.$$

As $t \to \infty$, $e^{-t/5} \to 0$, so $x \to \frac{200}{0+1} = 200$.

38 a)

$$\int \frac{15000}{P(1000-P)}\,dP = \int 1\,dt.$$

LHS:

$$\frac{15000}{P(1000-P)} \equiv \frac{A}{P} + \frac{B}{1000-P} \Rightarrow 15000 \equiv A(1000-P) + BP$$

$$\text{Let } P = 0: \quad 15000 = 1000A \Rightarrow A = 15$$

$$\text{Let } P = 1000: \quad 15000 = 1000B \Rightarrow B = 15$$

$$\therefore \int (\frac{15}{P} + \frac{15}{1000-P})\,dP = \int 1\,dt$$

$$\Rightarrow 15\ln|P| - 15\ln|1000-P| = t + c$$

$$t = 0,\ P = 4000: \quad 15\ln 4000 - 15\ln 3000 = 0 + c \Rightarrow c = 15\ln\frac{4}{3}$$

$$\therefore 15\ln|P| - 15\ln|1000-P| = t + 15\ln\frac{4}{3}$$

$$\Rightarrow \ln|P| - \ln|1000-P| = \frac{1}{15}t + \ln\frac{4}{3}$$

$$\Rightarrow \ln|\frac{P}{1000-P}| = \frac{1}{15}t + \ln\frac{4}{3}$$

$$\Rightarrow \frac{P}{1000-P} = -\frac{4}{3}e^{t/15}.$$

We must take the negative **RHS** as $P = 4000$ when $t = 0$, so for this condition to be satisfied both sides must be negative.

$$\Rightarrow P = -\frac{4000}{3}e^{t/15} + \frac{4}{3}Pe^{t/15}$$

$$\Rightarrow P - \frac{4}{3}Pe^{t/15} = -\frac{4000}{3}e^{t/15}$$

$$\Rightarrow P(1 - \frac{4}{3}e^{t/15}) = -\frac{4000}{3}e^{t/15}$$

$$\Rightarrow P = \frac{-(4000/3)e^{t/15}}{1-(4/3)e^{t/15}}$$

$$\Rightarrow P = \frac{-4000e^{t/15}}{3-4e^{t/15}} = \frac{4000e^{t/15}}{4e^{t/15}-3}$$

$$\Rightarrow P = \frac{4000}{4-3e^{-t/15}} \text{ so } a = 4 \text{ and } b = -3.$$

38 b)

As $t \to \infty$, $e^{-t/15} \to 0$, so $P \to \frac{4000}{4-3\times 0} = 1000$.

39 a)

$$u = 2x^{1/2} - 1 \Rightarrow \frac{du}{dx} = x^{-1/2} = \frac{1}{\sqrt{x}} \Rightarrow dx = \sqrt{x}\, du$$

$$u = 2x^{1/2} - 1 \Rightarrow x^{1/2} = \frac{1}{2}u + \frac{1}{2}$$

$$\therefore \int \frac{1}{2\sqrt{x} - 1}\, dx = \int \frac{1}{u} \times \sqrt{x}\, du = \int \frac{1}{u} \times (\frac{1}{2}u + \frac{1}{2})\, du$$

$$= \frac{1}{2}\int \frac{u+1}{u}\, du = \frac{1}{2}\int (1 + \frac{1}{u})\, du = \frac{1}{2}u + \frac{1}{2}\ln|u| + k$$

$$= \frac{1}{2}(2x^{1/2} - 1) + \frac{1}{2}\ln|2x^{1/2} - 1| + k$$

$$= x^{1/2} - \frac{1}{2} + \frac{1}{2}\ln|1 - 2x^{1/2}| + k$$

$$= \sqrt{x} + \frac{1}{2}\ln|1 - 2\sqrt{x}| + c \text{ where } c = k - \frac{1}{2}.$$

39 b)

$$\int \frac{1}{2\sqrt{x} - 1}\, dx = \int \frac{1}{50}t^{-2/5}\, dt$$

$$\Rightarrow \sqrt{x} + \frac{1}{2}\ln|1 - 2\sqrt{x}| = \frac{1}{30}t^{3/5} + c$$

$$x = 0,\ t = 0: \quad 0 + 0 = 0 + c \Rightarrow c = 0$$

$$\therefore \sqrt{x} + \frac{1}{2}\ln|1 - 2\sqrt{x}| = \frac{1}{30}t^{3/5}$$

$$\Rightarrow t^{3/5} = 30\sqrt{x} + 15\ln|1 - 2\sqrt{x}|$$

$$\Rightarrow t = (30\sqrt{x} + 15\ln|1 - 2\sqrt{x}|)^{5/3} \text{ as required.}$$

39 c)

$$x \geq 0 \text{ due to } \sqrt{x}$$

$$1 - 2\sqrt{x} > 0 \text{ due to } \ln|1 - 2\sqrt{x}|$$

$$1 - 2\sqrt{x} > 0 \Rightarrow \sqrt{x} < \frac{1}{2} \Rightarrow x < \frac{1}{4}$$

$$\therefore 0 \leq x < \frac{1}{4}.$$

40)

$$\int \cos 4y\, dy = \int 4xe^{2x}\, dx.$$

RHS:

$$\begin{aligned} u = 4x, \quad \frac{dv}{dx} &= e^{2x} \\ \frac{du}{dx} = 4, \quad v &= \frac{1}{2}e^{2x} \\ \therefore \int \cos 4y\, dy &= 2xe^{2x} - \int 2e^{2x}\, dx \\ &\Rightarrow \frac{1}{4}\sin 4y = 2xe^{2x} - e^{2x} + k \\ &\Rightarrow \sin 4y = 8xe^{2x} - 4e^{2x} + c \\ &\Rightarrow 4y = \arcsin(8xe^{2x} - 4e^{2x} + c) \\ &\Rightarrow y = \frac{1}{4}\arcsin(8xe^{2x} - 4e^{2x} + c) \\ &\Rightarrow y = \frac{1}{4}\arcsin(4e^{2x}(2x-1) + c). \end{aligned}$$

41)

$$\int y \sin 4y\, dy = \int (3x-1)^5\, dx.$$

LHS:

$$\begin{aligned} u = y, \quad \frac{dv}{dy} &= \sin 4y \\ \frac{du}{dy} = 1, \quad v &= -\frac{1}{4}\cos 4y \\ \therefore -\frac{1}{4}y\cos 4y + \int \frac{1}{4}\cos 4y\, dy &= \int (3x-1)^5\, dx \\ &\Rightarrow -\frac{1}{4}y\cos 4y + \frac{1}{16}\sin 4y = \frac{1}{18}(3x-1)^6 + c. \end{aligned}$$

42)

$$\int y^2 \sin 3y \, dy = \int 6\cos^2 2x \, dx.$$

LHS:

$$u = y^2, \quad \frac{dv}{dy} = \sin 3y, \qquad \frac{du}{dy} = 2y, \quad v = -\frac{1}{3}\cos 3y.$$

RHS:

$$\cos 2A = 2\cos^2 A - 1$$
$$\text{Let } A = 2x \Rightarrow \cos 4x = 2\cos^2 2x - 1$$
$$\Rightarrow \cos^2 2x = \frac{1}{2}\cos 4x + \frac{1}{2}$$
$$\therefore -\frac{1}{3}y^2 \cos 3y + \int \frac{2}{3} y \cos 3y \, dy = \int (3\cos 4x + 3)\, dx.$$

LHS:

$$u = \frac{2}{3}y, \quad \frac{dv}{dy} = \cos 3y, \qquad \frac{du}{dy} = \frac{2}{3}, \quad v = \frac{1}{3}\sin 3y.$$
$$\therefore -\frac{1}{3}y^2 \cos 3y + \frac{2}{9} y \sin 3y - \int \frac{2}{9}\sin 3y \, dy = \int (3\cos 4x + 3)\, dx$$
$$\Rightarrow -\frac{1}{3}y^2 \cos 3y + \frac{2}{9} y \sin 3y + \frac{2}{27}\cos 3y$$
$$= \frac{3}{4}\sin 4x + 3x + c.$$

43)

$$\frac{dN}{dt} = \frac{(3000 - N)(3000 + N)}{500}$$
$$\Rightarrow \int \frac{1}{(3000 - N)(3000 + N)}\, dN = \int \frac{1}{500}\, dt.$$

LHS:

$$\frac{1}{(3000 - N)(3000 + N)} \equiv \frac{A}{3000 - N} + \frac{B}{3000 + N}$$
$$\Rightarrow 1 \equiv A(3000 + N) + B(3000 - N)$$
$$\text{Let } N = 3000: \quad 1 = 6000A \Rightarrow A = \frac{1}{6000}$$
$$\text{Let } N = -3000: \quad 1 = 6000B \Rightarrow B = \frac{1}{6000}$$
$$\therefore \int \left(\frac{1}{6000(3000 - N)} + \frac{1}{6000(3000 + N)}\right) dN = \int \frac{1}{500}\, dt$$
$$\Rightarrow -\frac{1}{6000}\ln|3000 - N| + \frac{1}{6000}\ln|3000 + N| = \frac{1}{500}t + c$$
$$\Rightarrow \ln|3000 + N| - \ln|3000 - N| = 12t + A$$
$$N = 2000, \ t = 0: \quad \ln 5000 - \ln 1000 = 0 + A \Rightarrow A = \ln 5$$
$$\therefore \ln|3000 + N| - \ln|3000 - N| = 12t + \ln 5$$
$$\Rightarrow \ln\left|\frac{3000 + N}{3000 - N}\right| = 12t + \ln 5 \Rightarrow \frac{3000 + N}{3000 - N} = \pm 5e^{12t}.$$

Assuming that N can increase from 2000 up to 3000, both sides of the equation need to be positive.

$$\begin{aligned}
&\therefore \frac{3000+N}{3000-N} = 5e^{12t} \\
&\Rightarrow 3000+N = 15000e^{12t} - 5Ne^{12t} \\
&\Rightarrow N + 5Ne^{12t} = 15000e^{12t} - 3000 \\
&\Rightarrow N(1+5e^{12t}) = 15000e^{12t} - 3000 \\
&\Rightarrow N = \frac{15000e^{12t}-3000}{1+5e^{12t}} \Rightarrow N = \frac{15000-3000e^{-12t}}{e^{-12t}+5}.
\end{aligned}$$

As $t \to \infty$, $e^{-12t} \to 0$, so $N \to \frac{15000-3000\times 0}{0+5} = 3000$.

44)

Rate **out**: $0.1h^2$.

Rate **in**: 0.4.

$$\begin{aligned}
\frac{dV}{dt} &= 0.4 - 0.1h^2 \Rightarrow \frac{dV}{dt} = 0.1(4-h^2) \\
V &= 5\times 3\times h = 15h \Rightarrow \frac{dV}{dh} = 15 \\
\frac{dV}{dt} &= \frac{dV}{dh}\times\frac{dh}{dt} = 15\times\frac{dh}{dt} \\
\therefore 15\frac{dh}{dt} &= 0.1(4-h^2) \Rightarrow \int \frac{1}{4-h^2}\,dh = \int \frac{1}{150}\,dt \\
&\Rightarrow \int \frac{1}{(2+h)(2-h)}\,dh = \int \frac{1}{150}\,dt \\
\frac{1}{(2+h)(2-h)} &\equiv \frac{A}{2+h} + \frac{B}{2-h} \Rightarrow 1 \equiv A(2-h) + B(2+h) \\
\text{Let } h=-2: \quad 1 &= 4A \Rightarrow A = \frac{1}{4} \\
\text{Let } h=2: \quad 1 &= 4B \Rightarrow B = \frac{1}{4} \\
\therefore \int (\frac{1}{4(2+h)} + \frac{1}{4(2-h)})\,dh &= \int \frac{1}{150}\,dt \\
&\Rightarrow \frac{1}{4}\ln|2+h| - \frac{1}{4}\ln|2-h| = \frac{1}{150}t + c \\
t=0,\ h=0: \quad \frac{1}{4}\ln 2 - \frac{1}{4}\ln 2 &= 0 + c \Rightarrow c = 0 \\
\therefore \frac{1}{4}\ln|2+h| - \frac{1}{4}\ln|2-h| &= \frac{1}{150}t \Rightarrow t = \frac{75}{2}\ln|\frac{2+h}{2-h}| \\
\text{When } h=0.1, \quad t &= \frac{75}{2}\ln\frac{21}{19} = 3.75 \text{ seconds.}
\end{aligned}$$

45)

$$\int \frac{1}{y}\,dy = \int \frac{1}{8-2\sqrt{x}}\,dx.$$

RHS:

$$u = 8 - 2x^{1/2} \Rightarrow \frac{du}{dx} = -x^{-1/2} \Rightarrow dx = -x^{1/2}\,du$$

$$u = 8 - 2x^{1/2} \Rightarrow x^{1/2} = 4 - \frac{1}{2}u$$

$$\therefore \int \frac{1}{y}\,dy = \int \frac{1}{u} \times -x^{1/2}\,du$$

$$\Rightarrow \int \frac{1}{y}\,dy = \int \frac{1}{u} \times -(4 - \frac{1}{2}u)\,du$$

$$\Rightarrow \int \frac{1}{y}\,dy = \int (-\frac{4}{u} + \frac{1}{2})\,du$$

$$\Rightarrow \ln|y| = -4\ln|u| + \frac{1}{2}u + c$$

$$\Rightarrow \ln|y| = -4\ln|8 - 2x^{1/2}| + \frac{1}{2}(8 - 2x^{1/2}) + c$$

$$\Rightarrow \ln|y| = -4\ln|8 - 2x^{1/2}| - x^{1/2} + A$$

$$y = 1,\ x = 0: \quad 0 = -4\ln 8 + A \Rightarrow A = 4\ln 8$$

$$\therefore \ln|y| = -4\ln|8 - 2x^{1/2}| - x^{1/2} + 4\ln 8$$

$$\Rightarrow \ln|y| = 4\ln|\frac{8}{8 - 2x^{1/2}}| - x^{1/2}$$

$$\Rightarrow \ln|y| = \ln(\frac{4}{4 - x^{1/2}})^4 - x^{1/2}$$

$$\Rightarrow y = e^{\ln(4/(4-x^{1/2}))^4 - x^{1/2}}$$

$$\Rightarrow y = (\frac{4}{4 - x^{1/2}})^4 e^{-x^{1/2}} \Rightarrow y = \frac{256e^{-\sqrt{x}}}{(4 - \sqrt{x})^4}.$$

46)

$$\frac{1}{16}y^{-2}\frac{dy}{dx} = (3x+1)^{-3}$$

$$\Rightarrow \int \frac{1}{16}y^{-2}\,dy = \int (3x+1)^{-3}\,dx$$

$$\Rightarrow -\frac{1}{16}y^{-1} = -\frac{1}{6}(3x+1)^{-2} + c$$

$$y=2,\ x=1: \quad -\frac{1}{16}\times 2^{-1} = -\frac{1}{6}(3\times 1+1)^{-2} + c$$

$$\Rightarrow c = -\frac{1}{48}$$

$$\therefore -\frac{1}{16}y^{-1} = -\frac{1}{6}(3x+1)^{-2} - \frac{1}{48}$$

$$\Rightarrow y^{-1} = \frac{8}{3}(3x+1)^{-2} + \frac{1}{3} \Rightarrow \frac{1}{y} = \frac{8}{3}\times\frac{1}{(3x+1)^2} + \frac{1}{3}$$

$$\Rightarrow \frac{1}{y} = \frac{8}{3(3x+1)^2} + \frac{(3x+1)^2}{3(3x+1)^2} \Rightarrow \frac{1}{y} = \frac{8+(3x+1)^2}{3(3x+1)^2}$$

$$\Rightarrow \frac{1}{y} = \frac{8+9x^2+6x+1}{3(9x^2+6x+1)} \Rightarrow \frac{1}{y} = \frac{9x^2+6x+9}{27x^2+18x+3}$$

$$\Rightarrow \frac{1}{y} = \frac{3x^2+2x+3}{9x^2+6x+1} \Rightarrow y = \frac{9x^2+6x+1}{3x^2+2x+3}.$$